

Junior Cedric Tonga

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Research Interests: NLP • Multilingual NLP • Human-Centered NLP • Reasoning and Agents Systems • Evaluation of LLMs and Safety • NLP Applications (Education, Finance)

EDUCATION

- PhD Student in NLP, MBZUAI** Abu Dhabi, UAE
Agents systems, interpretability, Machine learning security / safety, advanced NLP Aug. 2025 – Aug. 2029 (expected)
- Master 2 MVA (Mathematics, Vision, Learning), ENS Paris-Saclay; CGPA: 4/4** Gif-sur-Yvette, France
Computer vision, Advanced learning for text and graph data, Time Series, ML, DL Oct. 2023 – Apr. 2024
- Master 1 in artificial intelligence, Université Paris-Saclay; CGPA: 3.7/4** Gif-sur-Yvette, France
Applied statistics, maths for data science, NLP, optimization, big data (Hadoop...), ML, DL Sept. 2022 – Apr. 2023
- Bachelor's degree in computer science, Faculty of sciences of Gabes; CGPA: 4/4** Gabes, Tunisia
Data structure and algorithmic, Probability and Statistic, Linear algebra, programming sept. 2019 – Jun. 2022

EXPERIENCES

- MBZUAI, NLP department** Abu Dhabi, UAE
Research associate II Oct. 2024 – Jul. 2025
- Supervisor: Dr Fajri Koto
 - Built a benchmark for cultural commonsense reasoning in Arabic.
 - Designed a prompt-based method to extract multilingual cultural knowledge graphs with LLMs.
 - Simulated student-tutor interactions with LLMs for multilingual math feedback
- EvidenceB, AI Research team** Paris, France
NLP research scientist intern Apr. 2024 – Aug. 2024
- Supervisors: Dr Pierre-Yves Oudeyer and Dr Benjamin Clement
 - Exploring the use of LLMs (GPT-4o and Llama-3-8B-instruct) as teachers to generate effective hints for students simulated through LLMs (GPT-3.5-turbo, Llama-3-8B-Instruct, Mistral-7B-instruct-v0.3) tackling math exercises designed for human high-school students, and designed using cognitive science principles.
 - Identified common student errors, developed prompts to aid self-correction, and compared teacher models performance using best prompt in generating pedagogically effective hints.
 - Results revealed that effective hints improved student model performance, especially GPT-3.5-turbo at lower temperatures. Llama-3-8B-Instruct as a teacher showed better overall performance than GPT-4o and Mistral-7B-Instruct showing decreased accuracy at higher temperatures.
 - **Paper accepted at NeurIPS 2024 FM-Assess Workshop (poster): Automatic Generation of Question Hints for Mathematics Problems using Large Language Models in Educational Technology.**
- Université Quebec à Montréal, CIRST lab** Montreal, Canada
NLP research intern Jun. 2023 – Aug. 2023
- Supervisors: Dr Marie-Jean Meurs and Dr Diego Maupomé
 - Conducted a study to assess the robustness of monolingual and multilingual language models to specific linguistic structures within the context of suicide prevention tools aimed at accommodating the cultural diversity of individuals in distress. Trained XLM-R, distiluse-base, and CamemBERT-base models on a dataset of French sentence pairs using the simple contrastive learning of sentence embeddings (SimCSE) method.
 - Results revealed that while pre-trained multilingual models initially performed well, post-training, monolingual models demonstrated superior performance over multilingual models.
 - **Work presented at the ACFAS congress in Ottawa in May 2024.**
- Digital Research Center of Sfax, Brain4ICT Team** Sfax, Tunisia
NLP research intern Feb. 2022 – May 2022
- Supervisor: Dr wael ouarda
 - Researched, collected, and cleaned product review data in Francamglais Cameroonian dialect from YouTube via Python script. Utilized BERT for sentiment analysis, achieving an 86% accuracy post fine-tuning. Deployed the model on a web application using Flask for user sentiment analysis.
 - **Paper accepted at IWCMC 2024: AfriDial: African Dialect Model based on Deep Learning for Sentiment Analysis.**

PUBLICATIONS

Tonga, J. C., Chen Liu, Iryna Gurevych, Fajri Koto (2025). LLMs as Cultural Archives: Cultural Commonsense Knowledge Graph Extraction. **EACL 2026**

Abdelrahman, S., **Tonga, J. C.**, Khalid, A., Saeed, A., Farah, A., Chatrine, Q., Karima, K., Sara, S., Yaser, A., Fajri, K. (2025). Commonsense Reasoning in Arab Culture. **ACL 2025**

Tonga, J. C., Clement, B., Oudeyer, P. Y. (2024). Automatic Generation of Question Hints for Mathematics Problems using Large Language Models in Educational Technology. **NeurIPS 2024 Workshop on Large Foundation Models for Educational Assessment (FM-Assess)**. Published by Proceedings of Machine Learning Research (PLMR)..

Tonga, J. C., Srivatsa, K. V. A., Maurya, K. K., Koto, F., Kochmar, E. (2025). Simulating Student-Tutor Interactions with LLMs for Multilingual Math Feedback.

Sassi, A., **Tonga, J.**, Poaty, S., Steve, S., Abakar Adjid, D. I., Cherif, M., Ouarda, W. (2024). AfriDial: African Dialect Model using Deep Learning for Sentiment Analysis. **International Wireless Communications and Mobile Computing (IWCMC) 2024**.

ACHIEVEMENTS & AWARDS

UAE government scholarship <i>Awarded a fully funded scholarship by the UAE government.</i>	Abu Dhabi, UAE <i>Aug.2025-Aug.2029</i>
G-Research Paris Quant Challenge 2024 <i>First Place, G-Research Paris Quant Challenge 2024 (team of 2)</i>	Paris, France <i>Jan. 2024</i>
SaclAI school excellence scholarship (MixtAI) <i>Paris-Saclay awards €10k to top AI Master's students for academic excellence</i>	ENS Paris-saclay, France <i>Sept. 2023 – Jun. 2024</i>
Globalink Research Internship fellowship-MITACS($\simeq$ \$10kCAD) <i>12-week internship program at a Canadian university.</i>	UQAM, Canada <i>May 2023 – Aug. 2023</i>
Idex international internship grants(IDEX) <i>internship bursary awarded to international interns on the basis of academic results</i>	Université Paris-saclay, France <i>May 2023 – Aug. 2023</i>
SaclAI school excellence scholarship (MixtAI) <i>Paris-Saclay awards €10k to top AI Master's students for academic excellence</i>	Université Paris-saclay, France <i>Sept. 2022 – Jun. 2023</i>
Research Excellence Award (resigned) <i>€12k in the first & €10k in the second year of EUR Msc.</i>	Université sorbonne paris-nord, France <i>Sept. 2022 – Sept. 2024</i>
Hatem Ben Taher Award <i>Best student in bachelor's degree in computer science of all 15 schools of the University of Gabes.</i>	FSG, Gabes, Tunisia <i>Jun. 2022</i>
Tunisian government scholarship <i>cooperation scholarship between the Cameroonian and Tunisian states.</i>	FSG, Gabes, Tunisia <i>Sept. 2019 – Aug. 2022</i>

SERVICE

Primary reviewer, COLING 2025, StarSEM 2025.

TEACHING

Teaching Assistant | Mohamed bin Zayed University of Artificial Intelligence Aug. 2025 - November 2025

- Teaching Assistant for the Master's course 'Mathematical Foundations': conducted TD sessions and graded student assignments.

TECHNICAL SKILLS

Languages: French, English

Programming Languages : Python (advanced), Java (prior experience), C/C++(prior experience), SQL(prior experience).

Developer Tools & framework: Git, Docker, VS Code, Spark, hadoop, Linux, Latex

Libraries: PyTorch, Numpy, Pandas, Scikit-learn, Matplotlib, Seaborn, NLTK, SentenceTransformers, transformers

Other : High-Performance Computing, Hugging Face